

Optimizing Multi-faceted Autoscaling For Kubernetes Microservices Using Model-based Reinforcement Learning

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What ?

Microservices on Kubernetes require dynamic resource allocation to handle fluctuating traffic. BUT...

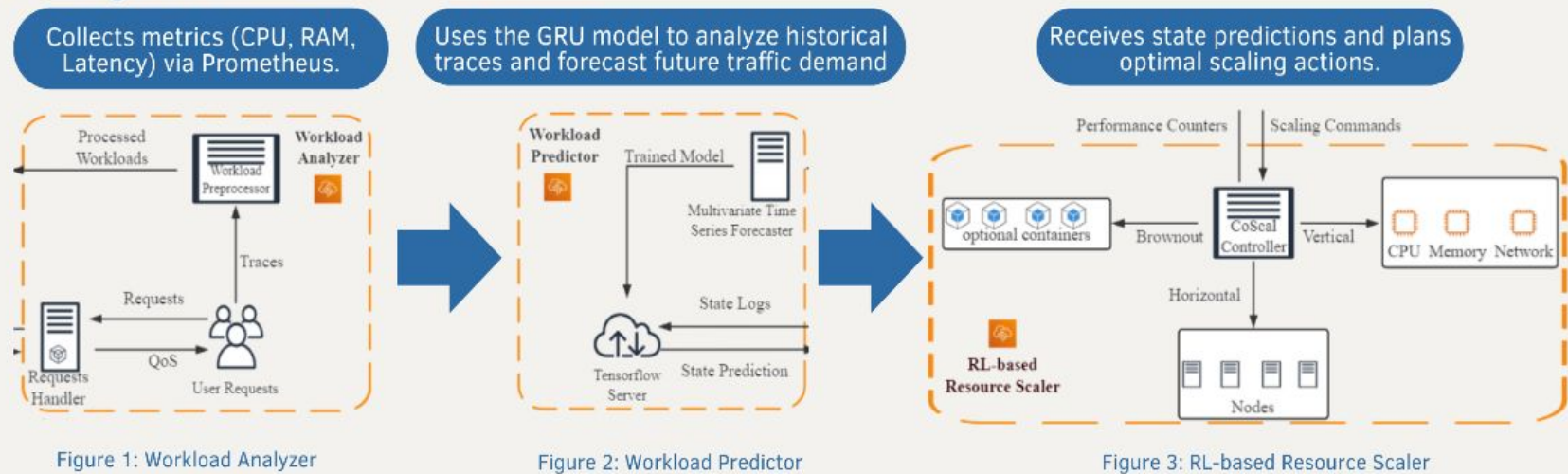
- Standard Kubernetes scaling (HPA/VPA) relies on reactive thresholds, causing latency during traffic bursts.
- Existing Reinforcement Learning (RL) solutions like CoScal are Model-free. They require long training times and risky "trial-and-error" in production.

Why ?

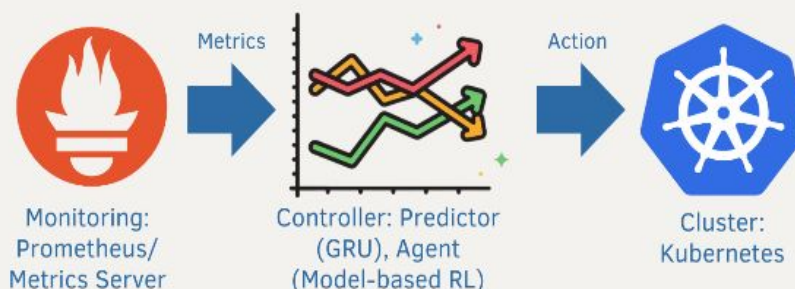
A novel autoscaling framework combining Multi-faceted Scaling (Horizontal, Vertical, Brownout) with Model-based RL to accelerate learning and ensure Stability, which help:

- Cost Efficiency: Minimizes cloud infrastructure costs by avoiding over-provisioning.
- SLA Assurance: Reduces response time violations during sudden load spikes.
- Practicality: Solving the "cold-start" problem of RL makes AI-driven scaling feasible for real-world DevOps workflows.

Proposed System Architecture



Overview



Methodology: Multi-faceted Actions



Brownout: Temporarily disables non-critical features.

Optimization: Model-based vs. Model-free

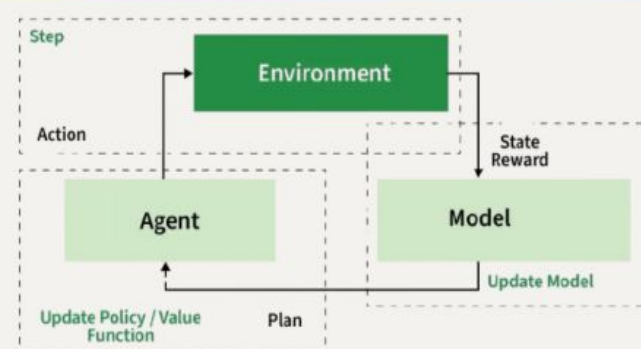


Figure 4: The Model-based RL Workflow

- **Model Learning**: The agent builds a virtual simulator representing the Kubernetes cluster dynamics.
- **Planning**: The agent trains its policy on the Simulated Model first. This allows safe "trial-and-error" without affecting the live production environment.

Expected Results

- **Convergence Speed**: The Model-based approach is expected to stabilize 30-40% faster than Model-free methods.
- **SLA Compliance**: Significant reduction in response time tail latency (P99) during bursty traffic scenarios.
- **Resource Utilization**: Higher efficiency in resource packing compared to default Kubernetes HPA.

Conclusion

This research addresses the critical trade-off between scaling agility and operational costs by proposing a robust Model-based Reinforcement Learning framework. By simulating actions prior to execution, the system establishes a vital "safety net" for autonomous scaling, effectively stabilizing decision-making in unpredictable production environments without relying on risky trial-and-error.

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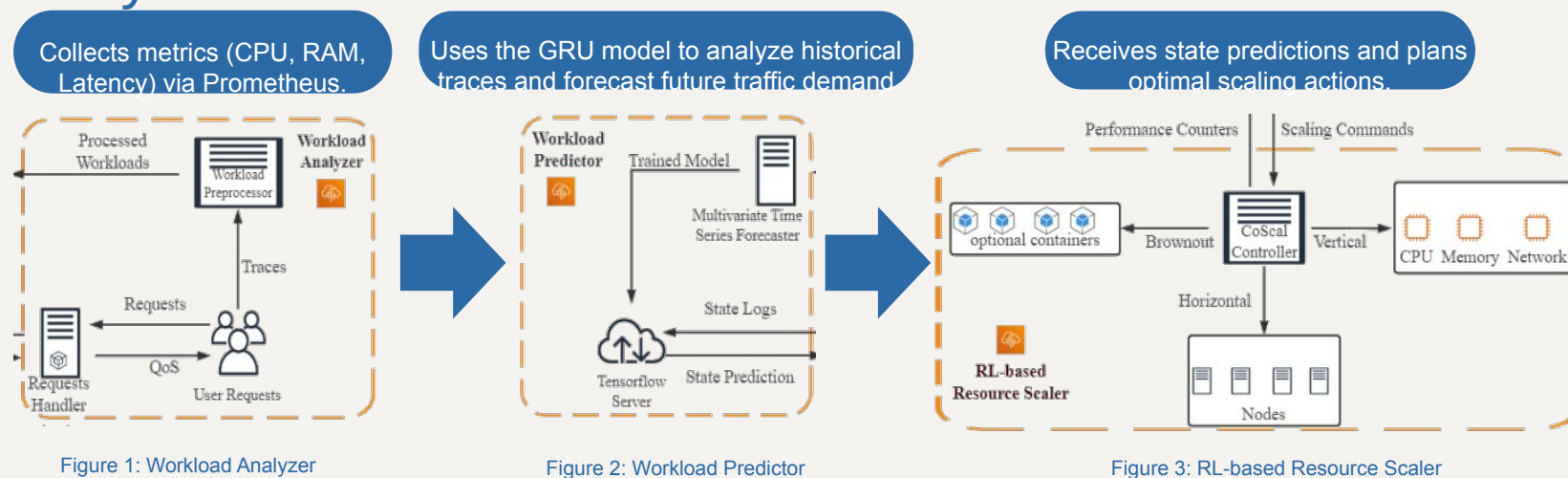
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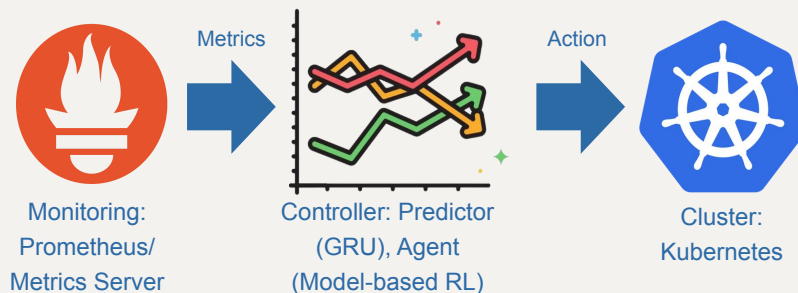
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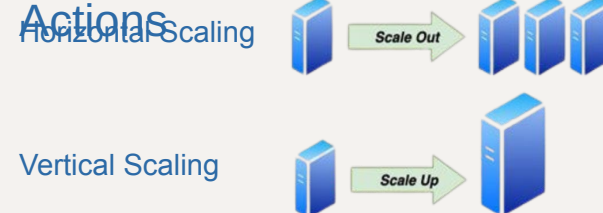
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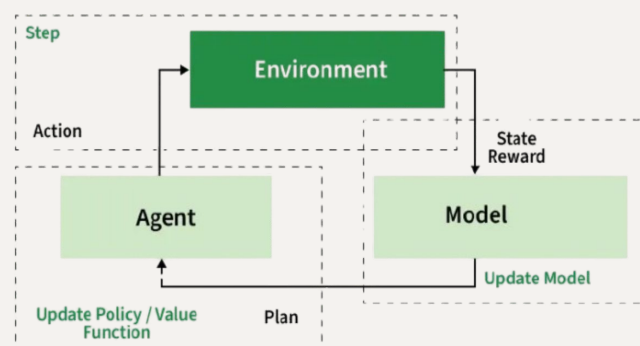


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